
SINGLE LEARNED NETWORK · one forward pass · no hand graph, no priors, no pruning stage
Reference
S_{ref}, P_t
Lean tokens + category embedding
RelHead
folded-in graph
s_{ij} attention bias
(pos + orient)
Rectified-flow DiT
v_θ (informative prior)
flat tokens · geo-bias
learned per-object heads
pose · wall-ness · ortho-ness · existence (keep)
pose + wall/ortho scores
Differentiable projection Π_θ
ONLY non-learned step
S_{tgt}
output
per-object token
state $x_\tau = (u, v, \cos\theta, \sin\theta)$
iterate $\tau: 0 \to 1$
lean cond: log-size · ref-pose · motif-head
learned category embedding
Linear $\to h$
flat tokens (no motif hierarchy)
$N\times$ DiT block (L=12)
adaLN(c) · scale/shift
Multi-Head Self-Attention
+ geo-bias + RelHead bias s_{ij}
per-object wall feature
nearest-wall dist · normal misalign
adaLN(c)
MLP (GELU)
residual · repeated $\times L$
τ time-embed
+ $g(P_t)$, room, z_{style}
RelHead
s_{ij} bias (pos+orient)
$c = \text{global}(\tau, g(P_t), \text{room}, z_{\text{style}})$
out adaLN + heads
pose · wall · ortho · keep
flow in world frame
Euler ODE
$x \leftarrow x + d\tau(v_\theta - \text{ramp} \cdot \nabla\Phi)$
50 steps · from prior x_0
K-step unrolled operator Π_θ (K=40)
anchor to proposal → nearest feasible point
DiT
v_θ
$\hat{x}_1$
$x^{(1)}$
$x^{(2)}$
$x^\star$
$\partial\mathcal{L}/\partial\theta$ back-prop through the projection
$x \leftarrow x - \eta\nabla_x [E_{\text{flush}} + E_{\text{ortho}} + E_{\text{col}} + w_a\|x - \hat{x}_1\|^2]$
CAPACITY (asset bank): shrink → existence head drops + smaller-asset substitution (−footprint)
grow → density-budgeted upsize + learned population (demand + PlacementNet)